CPSC 4420/6420 — Assignment 2

Tabular Q-learning in CliffWalking | Due: October 18, 2026 | Student: _____

Execution disclosure: Gymnasium was unavailable in the execution environment. Results and figures reported here were generated by the included local implementation of the standard 4×12 CliffWalking transition rules. The supplied Python code automatically uses official Gymnasium CliffWalking-v0 when installed. These local results should be re-run with Gymnasium for final environment verification.

A. Environment description (10 pts)

CliffWalking has a 4 × 12 grid, with start at the bottom-left (row 3, column 0), goal at bottom-right (row 3, column 11), and a 10-cell cliff between them. The goal is to reach the goal without stepping onto the cliff. The observation is a discrete integer in {0,…,47} encoding row × 12 + column. There are four discrete actions: up (0), right (1), down (2), and left (3). Actions normally move one cell; attempted moves beyond the border leave the agent in place. A regular step costs −1. Entering a cliff square returns the agent to start and gives −100, without ending the episode. Reaching the goal terminates the episode; a 200-step training/evaluation cap is imposed to prevent indefinite loops. States are already discrete, so no discretization is necessary. Baseline transitions are deterministic.

B. Q-learning algorithm (25 pts)

A Q table of shape 48 × 4 begins at zero. On each step, an ε-greedy action is chosen: a random action with probability ε, otherwise the first action attaining the maximum current Q-value. The Q-learning target is r for terminal or truncated transitions, and $r + \gamma \max_{a'} Q(s',a')$ otherwise. The update is $Q(s,a) \leftarrow Q(s,a) + \alpha[\text{target} - Q(s,a)]$. ε decays multiplicatively after every episode, subject to a floor. The agent is implemented from scratch; the environment API is used for reset and step, not for Q-learning.

C. Baseline configuration (10 pts)

Baseline: α=0.50, γ=0.99, ε initial=1.00, multiplicative ε decay=0.997, ε minimum=0.05, 3,000 episodes, max 200 steps/episode, random seed 42. Reward is cumulative reward per episode; the chart displays a 100-episode moving average. Training retains exploration, while reported policy evaluation disables it.

The final 100 training episodes average **-23.22** reward, whereas 100 greedy evaluation episodes average **-13.00**, reaching the goal in **13.00** steps. The gap reflects continuing ε-greedy exploration during training, including occasional cliff falls.

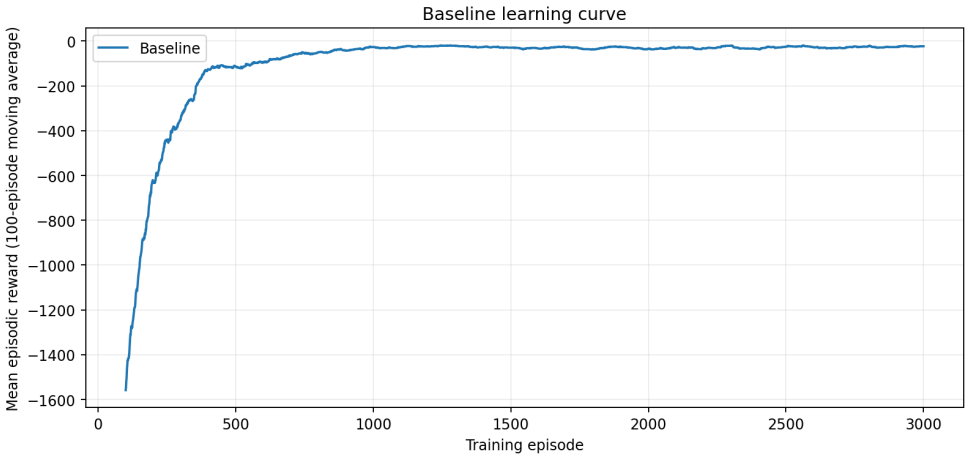


Figure 1. Baseline reward, smoothed over 100 training episodes.

D. Learning-rate experiment (15 pts)

Learning rates $\alpha \in \{0.1, 0.5, 0.9\}$; other baseline settings are fixed, including the random seed. Learning rate controls how aggressively new one-step targets replace past estimates. Compare each curve during early training and at convergence.

α	Final 100 training reward	Greedy mean reward	Success
0.1	-27.39	-13.00	100%
0.5	-23.22	-13.00	100%
0.9	-26.34	-13.00	100%

All tested learning rates reach the same final greedy shortest safe path in this deterministic setting. The lower learning rate retains more poor exploratory episodes near the end, while α=0.5 has the best final-100 training mean among the three. Large α can respond faster but amplify update variation during exploratory cliff falls.

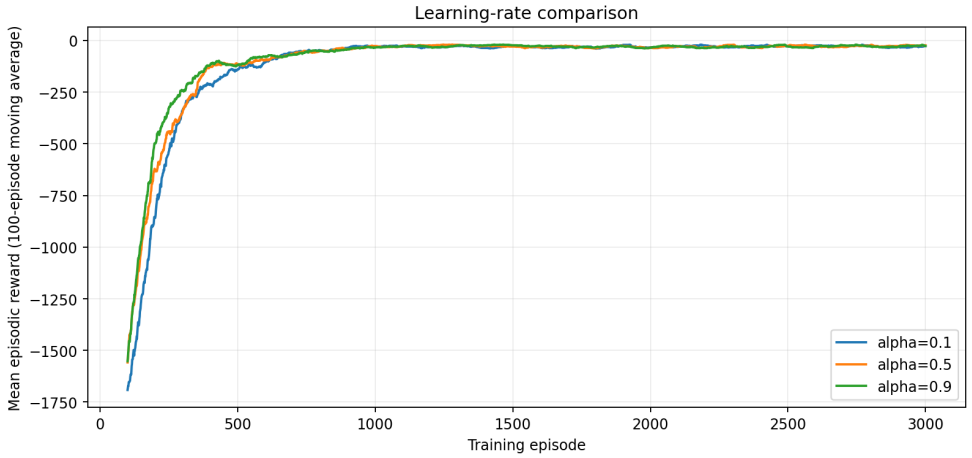


Figure 2. Learning-rate experiment; same remaining hyperparameters.

E. Discount-factor experiment (15 pts)

Discount factors $\gamma \in \{0.50, 0.90, 0.99\}$, with α, exploration, and training duration fixed. A larger γ attaches more weight to distant rewards, while a smaller γ emphasizes near-term costs. Here all three converge to the same greedy outcome, so the principal observed difference is in intermediate learning curves rather than final 100-episode evaluation.

γ	Final 100 training reward	Greedy mean reward	Success
0.5	-23.19	-13.00	100%
0.9	-23.22	-13.00	100%
0.99	-23.22	-13.00	100%

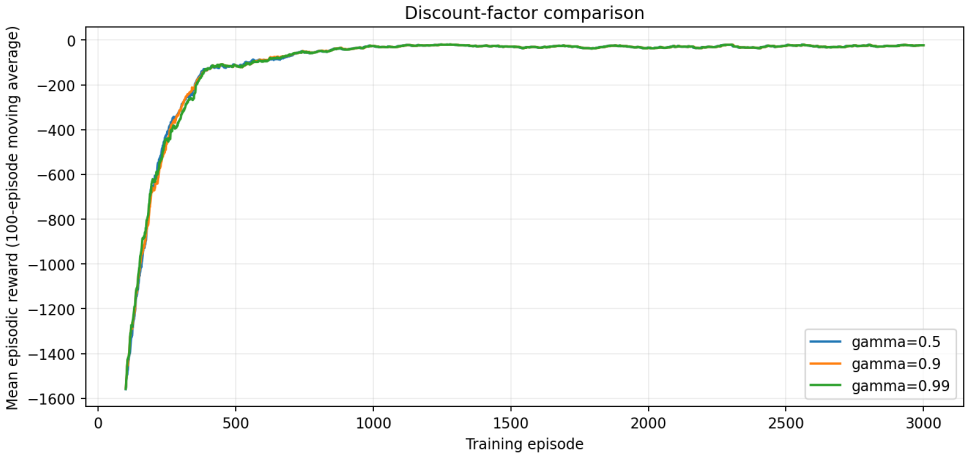


Figure 3. Discount-factor experiment.

F. Additional hyperparameter: epsilon decay (10 pts)

Tested multiplicative ε-decay values 0.99, 0.997, and 0.999 (ε initial 1.0; minimum 0.05). Faster decay reduces late random exploration but can limit early state coverage. Slower decay prolongs cliff risk in training. All three reach identical greedy-policy evaluation rewards, while 0.999 yields the worst last-100 training reward owing to continued exploration.

ε-decay	Final 100 training reward	Greedy mean reward	Success
0.99	-25.39	-13.00	100%
0.997	-23.22	-13.00	100%
0.999	-34.73	-13.00	100%

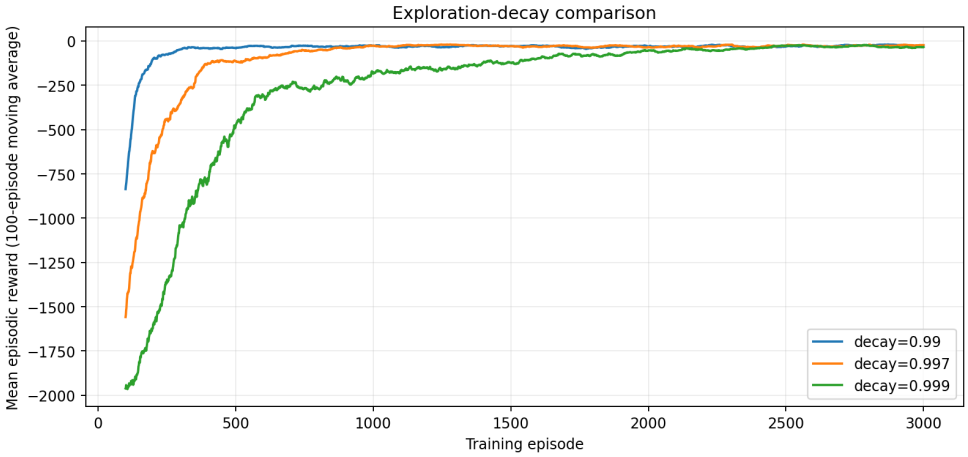


Figure 4. Exploration-decay experiment.

G. Evaluate the best agent (15 pts)

All ten baseline/parameter configurations were evaluated using 100 fresh episodes with ε=0. We selected the highest mean greedy evaluation reward, breaking ties by success rate and fewer average steps. Every configuration tied: mean reward –13.00, 100% success, and 13.00 steps. The baseline is retained as the first top-performing configuration, not because it significantly outperformed the others.

The 13-step route goes up from the starting square, travels across the safe row, and descends into the goal. Each step costs –1, yielding –13 total. The baseline and selected best policy therefore have identical reported metrics. No claim of a uniquely superior tuned hyperparameter set is supported by these results.

H. Extra credit: stochastic action corruption (10 pts)

A controlled uncertainty source was added: with probability p, the intended action is replaced uniformly by one of the other three valid actions. Three values p={0,0.10,0.25} were investigated. At each noise level a fresh Q table is trained for 3,000 episodes; the executed (possibly corrupted) action is used for learning. Evaluation is greedy except for injected noise. Separately, the originally selected p=0 policy is tested at each noise level without retraining.

Noise p	Retrained mean	Retrained success	Original-policy mean	Original success
0	-13.00	100%	-13.00	100%
0.1	-49.06	100%	-49.06	100%
0.25	-144.13	100%	-143.35	100%

As action corruption rises, mean reward becomes much worse despite every agent reaching the goal within the 200-step cap in this sample. This is because accidental cliff moves incur –100 and reset to start. Mean step counts also increase: 13.00 (p=0), 17.38 (p=0.10), and 27.31 (p=0.25) for freshly trained policies. The original deterministic policy retains 100% sampled completion, but incurs penalties of comparable magnitude under noisy execution. Consequently, success rate alone conceals a severe loss of robustness.

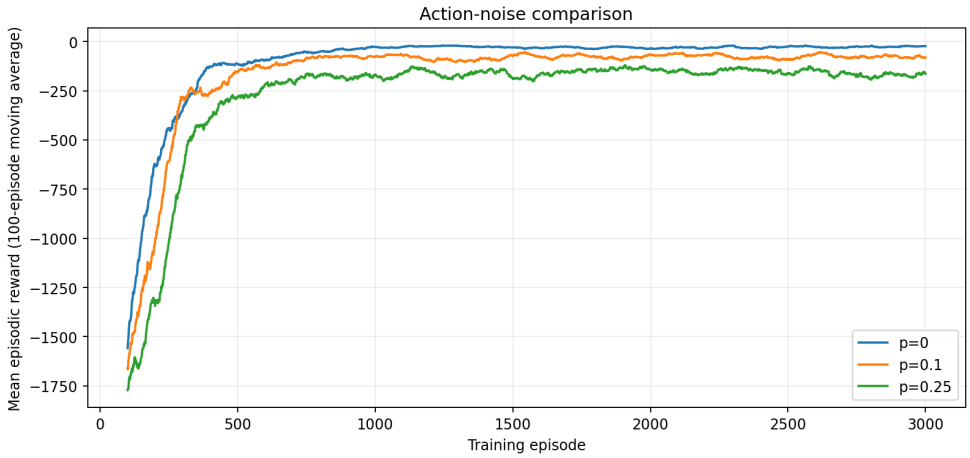


Figure 5. Training reward under differing action corruption levels.

Conclusions and reproducibility

Tabular Q-learning learned a 13-step, cliff-avoiding greedy route. Training reward is more volatile than final greedy evaluation because the training policy deliberately explores. In this small deterministic problem, widely varied learning rates, discount factors and epsilon decays all reach identical final greedy performance; convergence dynamics and training regret are more informative than the saturated success metric. Stochastic action errors reveal fragility despite high success rates. Limitations include one seed per configuration, a finite 100-episode evaluation sample, capped episode length, and local rather than Gymnasium-backed execution of the included figures. Future work should replicate runs with multiple seeds and report uncertainty intervals.

Repository: [Insert public GitHub repository URL after uploading project files]. Repository should include q_learning.py, requirements.txt, README.md, plots, metrics, and results/trained_agent.mp4. Only this PDF should be uploaded to Canvas.